

Notes on 60-degree-free subsets of the Leech minimal vectors

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Abstract

Let C be the 196560 minimal vectors of the Leech lattice Λ , and call $S \subseteq C$ *60°-free* if no two of its members subtend an angle of 60° . Such sets are the raw material of the recent lifting constructions that improved the kissing numbers in dimensions 25–31: each unit of $|S|$ is worth one or more spheres in the lifted configuration. This note reports what a sustained computational attack established about $\max |S|$ and about the structure of the largest known examples. The headline is a certified ladder $496 \leq \max |S| \leq 837$; the interesting part is *why* the ladder is stuck where it is, and a fairly complete structural picture of the 496-element sets, which turn out to live inside a Turyn decomposition $\mathbb{R}^{24} = B_1 \perp B_2 \perp B_3$ with $\Lambda \cap B_i \cong \sqrt{2}E_8$, and to be governed by a $\text{PG}(3, 2)$ of “channels” carrying 8×8 rook grids. Every claim below carries a verification label; the accompanying archive lets you re-check the ones marked as attached.

1 Scope and verification status

1.1 The object

Scale Λ so that minimal vectors have norm 32. Then $\langle v, w \rangle \in \{0, \pm 8, \pm 16, \pm 32\}$ for $v, w \in C$, $|C| = 196560$, and $\langle v, w \rangle = 16$ is exactly the 60° condition. Let G be the *conflict graph* on C with edges at inner product $+16$. Then G is 4600-regular, and $\text{Co}_0 = \text{Aut}(\Lambda)$ acts arc-transitively on it. A 60° -free set is precisely an independent set of G , and we write $\alpha = \alpha(G) = \max |S|$.

Throughout, Co_0 acts on C with rank 7: the orbitals on ordered pairs are the seven inner-product classes, with valencies

$$(v_{32}, v_{16}, v_8, v_0, v_{-8}, v_{-16}, v_{-32}) = (1, 4600, 47104, 93150, 47104, 4600, 1),$$

so C carries a commutative 6-class association scheme. This scheme is the substrate of everything in Section 2.

1.2 The three labels

Every claim in this note carries exactly one of the following.

[CERTIFIED-ATTACHED] An exact certificate for the claim is in the accompanying archive, together with a self-contained checker, or else the certificate is small enough that it is printed in full in this note and you can verify it by hand. Nothing is taken on trust from a solver.

[CERTIFIED-HELD] An exact certificate exists and has been re-verified, but is *not* in the archive — either because of bulk, or because reproducing it requires infrastructure (a GPU pass, a multi-hour exhaustive enumeration) that would not survive the trip. In every such case the method is described here in enough detail that a specialist can reconstruct the certificate independently. These are not attached; please do not read them as though they were.

[EVIDENCE] Search evidence, a solver verdict taken on trust, or a floating-point analysis. No proof is claimed. Negative results of budgeted searches live here, as do estimates.

Two remarks on the boundary. First, a solver’s **OPTIMAL** verdict, however trustworthy in practice, is [EVIDENCE] here unless an independent exact certificate was extracted and re-checked; this demotes several statements below that a reader might expect to be certified. Second, an exhaustive deterministic enumeration in exact arithmetic counts as certified even though its “certificate” is the enumeration itself; such claims are marked [CERTIFIED-HELD] and the completeness argument is stated.

1.3 Status of the headline claims

Claim	Label
$\alpha \geq 496$; an explicit set and 64 distinct maximal ones	[CERTIFIED-ATTACHED]
The 64 are pairwise inequivalent under Co_0	[CERTIFIED-HELD]
Every two-point method gives exactly $9360/11 = 850.\overline{90}$	[CERTIFIED-ATTACHED]
$\alpha \leq 837$ (three-point Terwilliger SDP, exact rational dual)	[CERTIFIED-ATTACHED]
$\omega(G) \leq 24$, by a two-line proof reproduced in full below	[CERTIFIED-ATTACHED]
$\omega(G) = 24$ exactly (explicit maximum clique; complete census)	[CERTIFIED-HELD]
Lifted clique cuts do not move the three-point bound	[CERTIFIED-HELD]
Four-point block dimensions; largest block 4107	[CERTIFIED-HELD]
Four-point bound would land near 820–830	[EVIDENCE]
Structure of the 496: Turyn blocks, duads, channels, rook grids	[CERTIFIED-HELD]
The five refuted structural hypotheses of Section 3.7	[CERTIFIED-HELD]
Turyn-symmetric optimum lies in $[496, 720]$; upper end	[EVIDENCE]
No 60° -free set of size 497 found by any search	[EVIDENCE]

1.4 Clean-machine verification

The archive holds six kinds of thing: the record 496-element set; the 64 sets of its plateau; the exact rational dual certificate for $\alpha \leq 837$; the orbital numbering that certificate is written in; a README; and one verifier script. The verifier needs nothing but Python 3 and NumPy — no network, no other data, no compiled code, no other package. It regenerates the extended binary Golay code from the generator polynomial of the cyclic $(23, 12, 7)$ code, checks its weight distribution, builds all 196560 minimal vectors from the three shapes, and recomputes the entire association scheme, the 148 orbitals and the structure constants of the 148-dimensional algebra from those vectors. Nothing in the archive depends on a stored copy of the lattice, and nothing is taken from the solver that produced the certificate.

It was run on a fresh empty directory outside every source tree, with the archive as the only input, using the distribution interpreter under an emptied environment. Excerpts, lightly elided:

```

$ /usr/bin/python3 -V                                -> Python 3.10.12
$ python3 -c 'import numpy; print(numpy.__version__, numpy.__file__)'
1.21.5 /usr/lib/python3/dist-packages/numpy/__init__.py
$ env -i PATH=/usr/bin:/bin /usr/bin/python3 verify.py --all

[1] regenerating the extended binary Golay code and the Leech minimal vectors
Golay code G24: 4096 codewords, weight distribution 0:1, 8:759, 12:2576,
16:759, 24:1; minimum distance 8; linear: OK
Leech minimal vectors: 196560 rows, all of squared norm 32, all distinct;
inner products lie in {-32,-16,-8,0,8,16,32}; valencies from a base
vector: -32:1, -16:4600, -8:47104, +0:93150, +8:47104, +16:4600, +32:1

[2] set file: S496.txt
S496.txt: |S| = 496, all rows are Leech minimal vectors, all distinct
Gram off-diagonal histogram over the 245520 ordered pairs:
{-32: 496, -8: 75008, 0: 95008, 8: 75008}
pairs at inner product 16 (60 degrees): 0 -> the set is 60-degree-free
antipodal (S = -S): yes
S496.txt: maximality sweep over all 196560 minimal vectors
free vertices (outside S, tightness 0): 0 -> S is MAXIMAL
minimum tightness outside S: 4, attained by 80 vectors
... 64 further set files, same verdicts ...

[3] dual certificate: sdp3_certificate.json
scheme: 7 classes with valencies (1, 4600, 47104, 93150, 47104, 4600, 1)
147 non-empty cells (i,j,k); the cell (3,3,3) of mutually orthogonal
triples splits into two orbitals of sizes 42240 and 924 per fixed y
-> D = 148 orbitals
structure constants c[u,s,t] of the 148-dimensional algebra recomputed
from the lattice (one base pair per orbital, 148 sweeps of 196560)
exact checks on c: support, row and column sums, |u|c^u_st =
|s|c^s_{ut^T}, unit, associativity: OK
the two PSD multipliers are given as sums of 138 and 132 rank-one terms
m_i * p_i p_i^T / 2^e with m_i >= 0 and p_i integral, so they are
positive semidefinite by construction -- no numerical PSD test is used
exact rational recomputation of h_q for all 43 variables: matches
bound = 100274998453663516018422097313608417887315219990267781\
39 / 1197262141301475670592458614961179049702139939205\
9392 = 837.5358661609
floor of the bound = 837 -> every 60-degree-free set of Leech
minimal vectors has at most 837 elements

RESULT ok=1 golay_words=4096 leech_vectors=196560 sets_checked=65
all_size_496=1 all_60deg_free=1 all_antipodal=1 distinct_sets=64
distinct_gram_histograms=1 free_vertices_total=0 all_maximal=1
min_tightness=4 distinct_tightness_spectra=5
certificate_bound_floor=837 elapsed_s=103.4

```

(The bound and the RESULT line are wrapped here for width; they are single lines in the transcript.)
Total: 103.4 seconds, exit status 0; peak resident memory on an equivalent run, 352 MB.

Everything else in this note is [CERTIFIED-HELD] or [EVIDENCE], and in particular the following are *not* in the archive: the maximum-clique data, the clique-cut certificate, the four-point orbit tables, and the structural computations of Section 3. Sections 2.4–2.6 and 3 say so at each point.

2 The bounds ladder, $496 \leq \alpha \leq 837$

2.1 The lower end: $\alpha \geq 496$

A 496-element 60°-free set is attached as an explicit integer matrix, together with a verifier. [CERTIFIED-ATTACHED] The set itself is not new: it is one of those found by Ma et al. (arXiv:2511.13391), and only the coordinate realisation differs here, by a permutation of the 24 coordinates.

Its Gram matrix has off-diagonal entries only in $\{-32, -8, 0, 8\}$, with multiplicities $496 : 75008 : 95008 : 75008$ over the $496 \cdot 495$ ordered off-diagonal pairs. In particular it contains no pair at $+16$ (independence) and, as a bonus that was not asked for, no pair at -16 either; the 496 antipodal pairs say $S = -S$. By shape, S consists of 258 vectors of type $(\pm 2^8, 0^{16})$, 232 of type $(\mp 3, \pm 1^{23})$ and 6 of type $(\pm 4, \pm 4, 0^{22})$. The second moment $\sum_{v \in S} vv^T$ has spectrum $\{576^8, 640^4, 704^8, 768^4\}$, i.e. $32 \cdot \{18^8, 20^4, 22^8, 24^4\}$: S spans $\mathbb{R}^{24}$ but is not a tight frame.

There are at least 64 such sets, each maximal; all 64 are attached and the verifier confirms they are distinct. [CERTIFIED-ATTACHED] That they are moreover pairwise inequivalent under Co_0 is established in Section 3.6 by a computation whose certificate is not attached: the verifier separates them only into five tightness-fingerprint classes. [CERTIFIED-HELD] *Falsified by:* a 497-element independent set. Nothing found one; see Section 2.7.

2.2 Every two-point method gives exactly 9360/11

This is the sharpest negative result in the note, and it is completely explicit, so we give the data in full. [CERTIFIED-ATTACHED]

The scheme's first eigenmatrix P — rows indexed by the seven common eigenspaces $E_0, \dots, E_6$, columns by the classes 32, 16, 8, 0, -8 , -16 , -32 — is

$$P = \begin{pmatrix} 1 & 4600 & 47104 & 93150 & 47104 & 4600 & 1 \\ 1 & 2300 & 11776 & 0 & -11776 & -2300 & -1 \\ 1 & 1000 & 1024 & -4050 & 1024 & 1000 & 1 \\ 1 & 350 & -704 & 0 & 704 & -350 & -1 \\ 1 & 76 & -320 & 486 & -320 & 76 & 1 \\ 1 & -10 & 16 & 0 & -16 & 10 & -1 \\ 1 & -20 & 64 & -90 & 64 & -20 & 1 \end{pmatrix}, \quad m = \begin{pmatrix} 1 \\ 24 \\ 299 \\ 2576 \\ 17250 \\ 95680 \\ 80730 \end{pmatrix},$$

where $m_r = \dim E_r$; note $\sum_r m_r = 196560$. Column 2 of P is the spectrum of the conflict graph: eigenvalues 4600, 2300, 1000, 350, 76, -10 , -20 with those multiplicities.

Ratio (Hoffman) bound. With $k = 4600$ and $\lambda_{\min} = -20$,

$$\alpha \leq \frac{N(-\lambda_{\min})}{k - \lambda_{\min}} = \frac{196560 \cdot 20}{4620} = \frac{9360}{11} = 850.\overline{90}, \quad \alpha \leq 850.$$

Association-scheme (Delsarte) LP. Maximise $\sum_i a_i$ over inner distributions with $a_{32} = 1$, $a_{16} = 0$, $a \geq 0$, $aQ \geq 0$ (where $Q = NP^{-1}$). Exact vertex enumeration over rationals gives optimum 9360/11, attained at the unique vertex

$$a^* = \left(1, 0, \frac{2944}{11}, \frac{3450}{11}, \frac{2944}{11}, 0, 1\right) \quad (\text{classes } 32, 16, 8, 0, -8, -16, -32).$$

The optimal dual is

$$\beta = \left(0, \frac{116}{231}, \frac{17}{77}, \frac{37}{462}, \frac{8}{385}, \frac{1}{462}, 0\right),$$

and one checks by hand that $F(i) := \sum_r \beta_r Q_{ir} = -1$ on *every* allowed class $i \in \{-32, -16, -8, 0, 8\}$, whence $\sum_i a_i \leq 1 + \sum_r \beta_r m_r = 9360/11$. This β is not an accident: it is exactly $\beta_r = (\lambda_r - \lambda_{\min})/(k - \lambda_{\min})$, i.e. Hoffman's multiplier vector. The LP cannot beat the ratio bound because its optimal dual *is* the ratio bound's certificate, and the sign constraints $a \geq 0$ are never active.

Lattice-free Delsarte LP. Take the allowed cosine set $T = \{-1, -\frac{1}{2}, -\frac{1}{4}, 0, \frac{1}{4}\}$ (i.e. forbid $\cos 60^\circ = \frac{1}{2}$) and run the classical spherical-code LP in $\mathbb{R}^{24}$ with Gegenbauer polynomials normalised by $G_k(1) = 1$. The exact optimum is 9360/11 at every truncation degree ≥ 4 ; the optimum is already attained by the degree-4 polynomial

$$f(t) = \frac{4992}{11} (t+1)t(t+\frac{1}{4})(t-\frac{1}{4}), \quad f(-\frac{1}{2}) = -\frac{234}{11} < 0, \quad f \equiv 0 \text{ on } T \setminus \{-\frac{1}{2}\},$$

with Gegenbauer coefficients $f_0 = 1$, $f_1 = 24$, $f_2 = 5083/77$, $f_3 = 4416/11$, $f_4 = 27600/77$ and $f(1)/f_0 = 9360/11$.

Theta functions and the ± 16 variant. Symmetry-reduced Lovász ϑ (drop $a \geq 0$) and Schrijver ϑ' (keep it) both equal 9360/11; so the sign constraints buy nothing. Forbidding *both* $+16$ and -16 — a strictly stronger combinatorial condition, which the record set happens to satisfy — also gives exactly 9360/11, both with and without the antipodality constraint $a_{-32} = 1$. Entrywise nonnegativity is likewise inert.

Why they all coincide, and what would change it. The multiplicities of $E_0, \dots, E_5$ are 1, 24, 299, 2576, 17250, 95680, which are exactly $\dim \text{Harm}_k(S^{23})$ for $k = 0, \dots, 5$. That is the 11-design property of C showing up: those six idempotents are literally the Gegenbauer constraints of degree ≤ 5 , so the scheme LP and the lattice-free LP see the same inequalities there. The seventh idempotent E_6 has dimension 80730, which is *not* a harmonic space — $\dim \text{Harm}_6(S^{23}) = 457470$, and E_6 is a proper Co_0 -constituent of it. It is the only constraint in the scheme LP that knows anything about the Leech lattice beyond its 11-design property, and *it is slack at the optimum*:

$$(a^*Q)_r = \left(\frac{9360}{11}, 0, 0, 0, 0, 0, \frac{2152800}{11} \right), \quad \beta_6 = 0.$$

So no two-point method can do better than 850 here, and improving the two-point rung is not a matter of finding a cleverer polynomial. A three-point relaxation is the first one that sees the lattice at all.

Falsified by: an error in the intersection numbers of the scheme (they are re-derived from the vector list, and the scheme axioms plus six standard identities are re-checked on load), or an arithmetic error in the dual above, which is short enough to check by hand.

2.3 The three-point bound: $\alpha \leq 837$

[CERTIFIED-ATTACHED] — the exact rational dual certificate is in the archive, and the checker there re-derives every coefficient matrix it needs from the regenerated lattice, taking nothing from the solver that produced it.

The relaxation. Fix $x \in C$. Let $\mathcal{O}$ be the set of orbits of $\text{Stab}_{\text{Co}_0}(x)$ on $C \times C$; equivalently, the Co_0 -orbits of ordered triples with a marked first point. There are exactly $D = 148$ of them, distributed over the pair classes as 7, 25, 27, 30, 27, 25, 7. For an independent set S define

$$\lambda_u = \#\{(a, b, c) \in S^3 : (a, b, c) \in O_u\}, \quad x_u = \frac{\lambda_u}{|S| \cdot \text{size}_u},$$

where $\text{size}_u = \#\{(y, z) : (x, y, z) \in O_u\}$. Imposing S_3 -symmetry ($x_u = x_{u\tau} = x_{\text{swap}(u)}$) leaves 43 free variables. The constraints are: $M_1 = \sum_u x_u B_u \succeq 0$; $M_2 = \sum_u (x_{\text{diag}(k(u))} - x_u) B_u \succeq 0$; $0 \leq x \leq 1$; $x_e = 1$; $\sum_k v_k x_{\text{diag}(k)} = |S|$; and $x_u = 0$ whenever a pairwise class of u is forbidden. The objective is $\sum_k v_k x_{\text{diag}(k)}$. Since $M_1 + M_2 = \sum_k x_{\text{diag}(k)} A_k$, the two-point constraint of Section 2.2 is implied and the bound can only improve on 850.

Symmetry reduction. The span of $\{B_u\}$, closed under left and right multiplication by the dual idempotents E_i^* and the class matrices A_k , has dimension exactly $148 = D$: the Terwilliger algebra $T(x)$ is the full centraliser algebra of the point stabiliser. This is worth stating carefully, because the span of the 147 nonempty triple-class matrices $E_i^* A_k E_j^*$ is one dimension smaller and is *not* closed under multiplication — the Terwilliger algebra is not spanned by the triple-class matrices, and the SDP must live in the full 148-dimensional algebra. The dimension count was done by row-reduction over a large prime field, which gives $\text{rank}_{\mathbb{F}_p} \leq \text{rank}_{\mathbb{Q}} \leq D$ and hence pins the value exactly.

The reason 148 exceeds 147 is a genuine piece of lattice structure: exactly one triple-class cell splits into two Co_0 -orbits, namely the most symmetric one, three mutually orthogonal minimal vectors, with $p_{00}^0 = 43164 = 42240 + 924$. So Co_0 is *not* transitive on ordered triples with prescribed pairwise classes, and the two orthogonal-triple flavours are exactly the information a two-point method cannot express.

The left-regular-representation trick. The B_u span a semisimple matrix $*$ -algebra $\mathcal{A}$ (the orbitals are closed under transposition), and $\mathbb{C}^C$ is a faithful $\mathcal{A}$ -module containing every irreducible $\mathcal{A}$ -module. Hence for $M \in \mathcal{A}$,

$$M \succeq 0 \text{ as a } 196560 \times 196560 \text{ matrix} \iff M \succeq 0 \text{ in the left regular representation of } \mathcal{A}.$$

Writing $c[w, u, t]$ for the structure constants and $\Delta = \text{diag}(\text{size})$, the matrix $G(x)[w, t] = \text{size}_w \sum_u x_u c[w, u, t]$ is symmetric with *integer* coefficient matrices whenever x is S_3 -symmetric, and $M(x) \succeq 0 \iff G(x) \succeq 0$, a 148×148 test. No numerical block-diagonalisation of the algebra is ever performed and no irrational number enters the exact side — which is precisely what makes exact rounding cheap.

Exact rounding. For rational $Y_1, Y_2 \succeq 0$ and rational multipliers $\mu, \rho \geq 0$, set $h_q = \text{obj}_q + \langle Y_1, G_q \rangle + \langle Y_2, G_q^y \rangle + (\mu^\top C y)_q - \rho_q$. Weak duality gives

$$|S| \leq h_e + \sum_{q \text{ free}} \rho_q + \sum_{q \text{ free}, q \neq e} \max(0, h_q),$$

every term an exact rational. Validity does not depend on the numerics at all: an inaccurate dual makes the right-hand side larger, never wrong.

The rounding of Y_1, Y_2 is the point worth copying. Rather than round a matrix and then test it for positive semidefiniteness, the solver's eigendecomposition $Y = \sum_i \lambda_i q_i q_i^\top$ is mapped into the G basis and each term is rounded *separately*: the eigenvalue to a nonnegative integer m_i and the vector to an integer vector P_i , both at a chosen power-of-two scale, so that $\hat{Y} = 2^{-T} \sum_i m_i P_i P_i^\top$ with $m_i \geq 0$. This is a nonnegative combination of rank-one terms, hence positive semidefinite *by construction*, and no PSD verification of a rounded matrix is ever required. The multipliers μ, ρ are rounded to dyadic rationals and clamped at ≥ 0 ; both sign conventions are tried and the better bound kept. The final inequality is evaluated in exact rational arithmetic.

The numbers. The relaxation's numerical optimum is 837.533238 (interior-point, status `optimal`); the exact rounded dual gives 837.5358661609..., a rational with denominator 2^{173} . Hence

$\alpha \leq 837.$

The rounding cost is 1.369, so the relaxation's own optimum is bracketed in $[837.5332, 837.5359]$; both ends floor to 837. Note carefully: 837 is a floor, not the three-point optimum. Sanity anchors from the identical code path: forbidding nothing returns $196565 \geq |C|$, and forbidding both $+16$ and $+8$ (i.e. $\cos \leq 0$) returns exactly 48, the orthoplex bound $2 \cdot 24$.

The decisive validity check. A three-point relaxation is easy to get subtly wrong — a normalisation slip produces a beautiful, meaningless number. The check that pins it down is that the *actual* 496-element set, and an independent 488-element set, are exactly feasible points of the same relaxation: their orbit densities satisfy $0 \leq x \leq 1$, $x_e = 1$, the forbidden coordinates vanish, $y \geq 0$, and both 148×148 blocks are positive semidefinite under an exact rational symmetric elimination — and the objective evaluates to 496 and 488 on the nose. Feeding $S = C$ returns 196560, again on the nose, with the second block identically zero as the derivation predicts. A wrong normalisation would show up immediately as an *infeasible* known set. Deliberately perturbing one coordinate of the 496's density vector does make the exact PSD test fail.

Falsified by: an independent set of size 838; or an error in the orbital structure constants, which are re-derived from the vector list and re-checked at load.

2.4 Clique cuts are provably inert

$\omega(G) = 24$, **exactly.** The upper bound is [CERTIFIED-ATTACHED] and needs no computation. If $v_1, \dots, v_k$ are minimal vectors with $\langle v_i, v_j \rangle = 16$ for $i \neq j$, their Gram matrix is $16(I_k + J_k)$, whose eigenvalues are $16(k+1)$ once and 16 with multiplicity $k-1$. This is positive definite, so the v_i are linearly independent in $\mathbb{R}^{24}$ and $k \leq 24$. That is the whole argument, and it uses no sphere-packing input whatever.

Equality is [CERTIFIED-HELD]: an explicit 24-clique is held, not attached. Its Gram matrix is $16(I+J)$ (re-checked in integer arithmetic here), its sum s satisfies $|s|^2 = 9600$ and $\langle v, s \rangle = 400$ for every member, the 24 vectors are a basis of $\mathbb{R}^{24}$, and by shape they are three vectors of type $(\pm 4, \pm 4, 0^{22})$ and 21 of type $(\pm 2^8, 0^{16})$. Reconstructing one is a short branch-and-bound in the common neighbourhood of an edge.

A warning for anyone re-deriving this: the natural family $\{4(e_1 + e_i)\}_{i=2}^{24}$ is a 23-clique and is *maximal*, because the only real vector at inner product 16 with all of it is $(3, 1^{23})$, whose coordinate sum $26 \not\equiv 4 \pmod{8}$ puts it outside Λ . Maximal is not maximum here. An exhaustive branch-and-bound in the common neighbourhood of an edge (891 vertices, 336-regular) confirms $\omega = 24$, and a complete enumeration gives the maximal-clique sizes as exactly $\{8, 12, 15, 17, 23, 24\}$, with 5,028,032,232 maximal cliques through a single edge — the two independent census routes (through an edge, through a triangle) agree exactly. [CERTIFIED-HELD] for the census.

The cuts, and why they do nothing. For a pair (a, b) of non-conflicting vectors and a clique K , the valid pair-conditioned liftings are

$$\sum_{v \in K} x_{u(a,b,v)} \leq x_{\text{diag}(i)}, \quad \sum_{v \in K} (x_{\text{diag}(\text{class}(a,v))} - x_{u(a,b,v)}) \leq 1 - x_{\text{diag}(i)},$$

with $i = \text{class}(a, b)$. Adding 225 such cuts built from maximum cliques leaves the exact bound at 837; adding 437 triangle cuts leaves it at 837; adding all 662 leaves it at 837. The maximum violation at the uncut optimum, scanned over 1430 generated cuts, is -2.5×10^{-14} : every cut is slack. Both the 496 and the 488 remain exactly feasible under all of them (80 of the 662 tight, the rest strict), which is the validity gate — a deliberately corrupted cut is caught by the 496. [CERTIFIED-HELD]

The structural reason is a one-line estimate, and it is the useful part of this subsection. The three-point optimum behaves like a near-product pseudo-density with one-point density $\sigma = 837.53/196560 \approx 1/234.7$. Conditioned on $(a, b) \in S$, the third-point density at any single vertex is $O(\sigma)$, so the left-hand side of the first cut is about $|K|\sigma x_{\text{diag}(i)}$ and the cut binds only once $|K| \gtrsim 1/\sigma \approx 235$. But $\omega(G) = 24$ — an order of magnitude short. Concretely, at the optimum the tightest instance uses 12% of its right-hand side. Entrywise nonnegativity (ϑ^+ strength) is already implied by the baseline, because the B_u are 0/1 matrices with disjoint supports. Clique structure alone could never push a vertex-transitive bound below $|C|/\omega = 8190$ anyway.

Conclusion: at three points, the remaining lever is not combinatorial sparsification. It is either four-point terms or the split orthogonal-triple cell used directly.

2.5 Four-point scoping: priced, not run

The natural next rung is the level-3 Lasserre / four-point analogue. The orbit data one level down is complete. [CERTIFIED-HELD]

For each pair class i , the four-point PSD blocks live in the centraliser algebra of $\text{Stab}(x, y_i)$, whose dimensions are

$$\dim T(x, y_i) = 148, 1893, [2971, 2972], [4097, 4107], [2971, 2972], 1893, 148$$

for classes $-32, -16, -8, 0, 8, 16, 32$. The two anchors (148, for the classes where the stabiliser degenerates to $\text{Stab}(x)$) and the ± 16 classes are exact; the five bracketed cells have rigorous lower ends from GPU-computed five-point histogram invariants (a Stab-invariant, hence a lower bound on the orbit count) and rigorous upper ends from explicit stabiliser labels carrying group-element witnesses (hence an upper bound). An independent re-run with three times as many stabiliser elements reproduced the label counts exactly, which suggests the upper ends are the truth and the histogram is simply too coarse there. There are 14133 orbits of ordered 4-tuples, of which 6183 are admissible (no conflicting pair) and 5721 have four distinct points; after S_4 -symmetrisation the variable count is bracketed in $[258, 6183]$ and is realistically 240–300. Variables are a non-issue.

The blocks are the issue. A relaxation in the style of Section 2.3 has 17 PSD blocks (3 membership blocks $\times$ 5 base-pair classes, plus 2 diagonal), the largest 4107×4107 , with $\sum d^2 = 1.1 \times 10^8$ and $\sum d^3 = 3.9 \times 10^{11}$. The cone scalar count $\sum d(d+1)/2 = 5.7 \times 10^7$ puts interior-point iterates and scaling work matrices at roughly 7 GB *before* the SDP data and the KKT factors, which is why this needs a 64–128 GB workstation rather than a laptop. Against the three-point solve this is a factor 28 in block dimension and about 2×10^4 in flops. The exact-certificate machinery survives — the left-regular-representation trick still applies at dimension 4107 — but the dense rational contractions of the three-point path do not, and would have to be rewritten as sparse integer contractions; separately, the 496’s quadruple distribution (the feasibility anchor that made the three-point bound trustworthy) is a $496^4 \approx 6 \times 10^{10}$ label-lookup problem that needs its own engineering. Realistic budget: two to four weeks including that rewrite.

Expected movement, and this is an estimate rather than a claim: the two-point to three-point step was $850.9 \rightarrow 837.5$, or -1.6% ; a comparable relative step lands near 820–830. The Gijswijt–Mittelman–Schrijver experience with quadruple distances in the binary-code setting suggests a few percent is the right order, occasionally more when the extremal structure is rigid. Nothing in the slack structure suggests a collapse toward 496. [EVIDENCE]

It was not run for exactly one reason: the largest block is 4107, and the available machine has 16 GB. Nothing was attempted and abandoned.

2.6 What the 837 optimum looks like

This is a floating-point analysis of the relaxation’s optimum, so [EVIDENCE] — but it is the most suggestive number in the note.

The optimum is massively degenerate: the two 148×148 PSD blocks have ranks 28 and 68; 65 of the 148 y -inequalities are tight; 11 of the 27 free non-identity orbit variables sit at zero. The optimal pseudo-set is antipodal ($x_{\text{diag}}(-32) = 1$, achieved and not imposed). And — without being told to — it puts *zero* mass on both +16 and −16. The relaxation discovers the ± 16 -avoidance of the record set on its own, even though Section 2.2 showed that no two-point method can see the −16 prohibition at all.

Its pair densities against the record’s, class by class:

class	optimum	the 496	ratio
± 32	1.000000	1.000000	1.000
± 16	0	0	—
± 8	0.005587	0.003210	1.740
0	0.003319	0.002056	1.614

The ratios bracket $837/496 = 1.6875$: the optimum looks like the 496 inflated by about 1.69. That the bound is pinned by a large active set of genuinely three-point constraints, rather than by one slack inequality, is the same conclusion Section 2.4 reached independently.

2.7 Search evidence against 497

All [EVIDENCE], and included because negative results save time.

- The record set is swap-optimal for every removal size $k \leq 12$, exhaustively: a complete depth-first search over unions of conflict sets (complete because every improving swap is witnessed by such a union) with exact bitset independent-set solving finds no improving move, and the only equal-size moves of size ≤ 12 are four disjoint (12, 12) plateau moves. Any improvement must remove at least 13 vectors. This one is an exhaustive exact computation, so [CERTIFIED-HELD] rather than [EVIDENCE].
- Two 30-minute production runs of a GPU iterated-local-search driver, 242 million iterations, best = 496 and flat. The census of distinct 496-element sets *saturates* at $4160 = 65 \times 64$ within about six minutes: every chain that reaches a 496 is trapped in its plateau component, and diversity comes only from group images of the seed, not from more chains or more time.
- A sweep over 95 subgroups of Co_0 — cyclic groups of all element orders with and without signs, Sylow subgroups, several Mathieu groups, 2^{12} and its overgroups, $\text{PSL}(2, 23)$, dihedral groups, random conjugates — maximising over unions of orbits. Only the record’s own stabilisers reach 496. The best union under any other group is 402. 23 of the rows finished branch-and-bound to optimality. Any group containing 2^{12} or a Sylow 2-subgroup caps at 48, one frame.

3 Structure of the 496-element sets

This section is the part most likely to be reusable. Everything in it is an exact computation with in-run assertions — integer and rational arithmetic throughout, including Hermite and Smith normal forms and exact short-vector enumeration — and is therefore [CERTIFIED-HELD] unless stated otherwise; the vector lists it starts from are attached, so any individual claim can be re-derived from the archive by a reader willing to write the code.

3.1 The Turyn decomposition

The setwise stabiliser of the record set in Co_0 has order 8 and is elementary abelian 2^3 . Its eight elements have traces $\{+24, +8, +8, +8, -8, -8, -8, -24\}$; the three trace-(+8) involutions t_1, t_2, t_3 satisfy $t_i t_j = -t_k$ (so no two of them generate a Klein group containing the third). Define B_i to be the (−1)-eigenspace of t_i . Then

$$\mathbb{R}^{24} = B_1 \perp B_2 \perp B_3, \quad \dim B_i = 8.$$

For each i , the Gram matrix of $\Lambda \cap B_i$, divided by 16, is an 8-dimensional even integral matrix of determinant 1; by uniqueness of the even unimodular lattice in dimension 8,

$$\Lambda \cap B_i \cong \sqrt{2} E_8,$$

confirmed independently by exact short-vector enumeration, which finds 240 vectors of norm 32 and 2160 of norm 64 in each block. The glue index is

$$[\Lambda : \sqrt{2}E_8 \oplus \sqrt{2}E_8 \oplus \sqrt{2}E_8] = 4096 = 2^{12},$$

with elementary abelian quotient (Smith form $[2^{12}]$). This is the Turyn construction of Λ from three copies of E_8 . Pairwise, $\Lambda \cap (B_j \oplus B_k) = \text{Fix}(t_i)$ is a 16-dimensional even lattice of determinant 2^8 with kissing number $4320 = 3840 + 2 \cdot 240$, i.e. the Barnes–Wall lattice BW_{16} , and $[\text{Fix}(t_i) : \sqrt{2}E_8 \oplus \sqrt{2}E_8] = 16$ with Smith form $[2, 2, 2, 2]$.

The stabiliser is thus concretely the *blockwise sign group*: writing $v = v_1 + v_2 + v_3$ with $v_i \in B_i$, the eight maps $v \mapsto \varepsilon_1 v_1 + \varepsilon_2 v_2 + \varepsilon_3 v_3$, $\varepsilon \in \{\pm 1\}^3$. Its elements are not monomial in the standard Leech coordinates.

3.2 Monads, duads, triads

Classify $s \in C$ by the multiset of block norms $(|s_1|^2, |s_2|^2, |s_3|^2)$. Only three patterns occur, because 8 and 24 are not norms of $\sqrt{2}E_8$:

name	pattern	count	in the record set
monad	$(32, 0, 0)$ — a $\sqrt{2}E_8$ minimal vector	$3 \times 240 = 720$	0
duad	$(16, 16, 0)$ — norm split $16 + 16$ across two blocks	$3 \times 3840 = 11520$	496
triad	$(16, 8, 8)$ — split across all three blocks	$3 \times 61440 = 184320$	0

$720 + 11520 + 184320 = 196560$. (This census was re-derived here from a Turyn triple recovered independently of the record’s stabiliser, and agrees.)

The record lives entirely in the duads. All 496 vectors are duads, split $208/144/144$ over the three block pairs — which are exactly the fixed-vector counts of t_1, t_2, t_3 on S , since a duad supported on $\{B_j, B_k\}$ is fixed by t_i . So the record uses 496 of 11520 duads, about 4.3%. All 64 sets of the plateau family are duad sets on the *same* three blocks with the *same* $(208, 144, 144)$ split, as are the 80 near-miss vertices of Section 3.6 (split $48/16/16$).

There is a pleasing counterpoint. The maximum clique of G from Section 2.4 consists of 24 *triads* — zero monads, zero duads. Duads carry the big independent sets; triads carry the big cliques; the two extremal structures of G occupy disjoint strata of the Turyn decomposition.

Under the blockwise sign group the 196560 minimal vectors fall into 360 orbits of size 2 (the monads), 2880 orbits of size 4 (the duads, each of the form $\{\pm x, \pm y\}$ with $x \perp y$), and 23040 orbits of size 8 (the triads). *Every* triad orbit contains an internal conflict, so no triad orbit can appear in *any* set invariant under the blockwise 2^3 . That leaves 3240 usable orbits of total weight 12240 — and the record is 124 of them.

3.3 Channels: a PG(3, 2)

Take a duad orbit $O = \{\pm x, \pm y\}$ supported on blocks $\{B_j, B_k\}$. Then $x + y$ and $x - y$ are two vectors of norm 64, one in each block, and they are congruent mod 2Λ . Their common class in $\Lambda/2\Lambda$ is the *channel* of the orbit.

Per block, the 2160 norm-64 vectors fall into 135 classes of 8 antipodal pairs each, and each class is an orthogonal frame of that 8-dimensional block — the 135 frames of E_8 . Of these, each pair of blocks shares exactly 15 channels, and — this is the fact that organises everything — *the same 15 for all three pairs*. Together with 0 they are closed under addition in $\Lambda/2\Lambda$ and span a 4-dimensional $\mathbb{F}_2$ -space. So:

The 15 channels are the 15 points of a PG(3, 2): 15 points, 35 lines $\{c, c', c \oplus c'\}$, 15 planes, and maximum caps of size 8 (the affine complements of the planes).

The rook grid. A *cell* is a (block-pair, channel) combination; there are $3 \times 15 = 45$ of them. Within the cell $(\{B_j, B_k\}, c)$, take as rows the 8 antipodal line pairs of channel c ’s frame in B_j and as columns the 8 in B_k ; each of the 64 positions is one duad orbit. Check: $45 \times 64 = 2880$ orbits, $\times 4 = 11520$ duads.

Two orbits in the same channel conflict if and only if they share a line: sharing a row or column gives inner product $(\pm 64 + 0)/4 = \pm 16$, which hits $+16$ for one of the sign choices, while distinct rows and distinct columns give 0. So within a cell, an independent set is a *non-attacking rook placement* — a partial permutation matrix, at most 8 orbits.

Per-channel independence is a matching condition. Now stop fixing the pair. A single channel c has 8 lines in each of the three blocks; a duad orbit in channel c is exactly an edge joining a line in one block to a line in another. The $3 \times 64 = 192$ orbits of channel c are precisely the 192 edges of the complete tripartite graph $K_{8,8,8}$, and independence within the channel is exactly the condition that the chosen edges form a *matching*. Since $K_{8,8,8}$ has 24 vertices, a channel carries at most 12 orbits, with equality exactly for a perfect matching.

The two caps together give $15 \times 12 = 180$ orbits, i.e. weight 720 — which is the ceiling of Section 4.4, and the reason it is hard to beat. The residual, and the genuinely hard part, is cross-channel conflict: two orbits in the same pair but different channels conflict iff *both* their row lines and their column lines are at 60° in their respective blocks; two orbits in different pairs and different channels *never* conflict; and monad-duad conflicts factor entirely through the plane geometry (a monad’s 14 conflicting lines lie in its own block, two in each of exactly 7 channels, and those 7 always form a plane of $\text{PG}(3, 2)$). Per block, both the 120 channel lines and the 120 monad lines under 60° -adjacency form the strongly regular graph $\text{srg}(120, 56, 28, 24)$ with spectrum $\{56^1, 8^{35}, (-4)^{84}\}$ — the E_8 root-line graph, whose ratio bound $120 \cdot 4/60 = 8$ is exactly why “at most 8 per block” keeps recurring.

3.4 The usage pattern $496 = 4 \cdot (3 \cdot 12 + 9 \cdot 8 + 2 \cdot 8)$

The record is a union of 124 orbits of the blockwise sign group, each of size 4. Its 45-cell usage profile is:

channel group	# channels	orbits each	profile over the pairs	total
frame (perfect $K_{8,8,8}$ matching)	3	12	$4 + 4 + 4$	36
one-pair (full 8-permutation)	9	8	$8 + 0 + 0$	72
half	2	8	$0 + 4 + 4$	16
empty	1	0	$0 + 0 + 0$	0

so $124 = 3 \cdot 12 + 9 \cdot 8 + 2 \cdot 8$ and $|S| = 4 \cdot 124 = 496$. Cell-wise: of the 45 cells, 9 carry a full 8-permutation, 13 carry 4, and 23 are empty; $9 \cdot 8 + 13 \cdot 4 = 124$. Per pair the totals are 52, 36, 36 orbits, i.e. 208, 144, 144 vectors, matching the block-support split.

Two geometric facts about this profile. The seven channels *unused* by the 208-pair, together with 0, form a hyperplane of the channel space — so that pair is supported on exactly a maximum cap of $\text{PG}(3, 2)$; the corresponding sets for the other two pairs are not subspaces. And the three frame channels span a plane, which contains all seven “special” channels (the three frames, the two half channels, the empty one, and one of the eight-permutation channels); the remaining eight channels form the complementary maximum cap and are all plain one-pair channels. The three frame channels are not collinear.

For scale: the record’s line budget per block is 72, 88, 88 of 1080 available antipodal pairs. It is not remotely space-constrained; it is constrained by the cross-channel interactions.

3.5 The three perfect channels are the three frames

The orthogonality graph on the 248 antipodal classes of the record — adjacency at inner product 0 — has 2718 maximal cliques, with size histogram $\{8:1280, 10:800, 12:448, 16:171, 18:4, 20:8, 22:4, 24:3\}$. The three 24-cliques are pairwise disjoint frames, covering 72 of the 248 classes, and they are exactly the three perfect-matching channels: a perfect matching of $K_{8,8,8}$ in channel c uses all 24 lines of c , its 12 orbits carry 48 vectors = 24 antipodal classes, and those classes are pairwise orthogonal because distinct matching edges share no line and cross-channel dots vanish. So *perfect channel* $\iff$ *24-clique in the orthogonality graph* $\iff$ *frame*.

The three frames are not interchangeable as lattice objects: each generates a rank-24 sublattice of index 2^{24} , but two of them have Smith form $[2^4, 4^{10}]$ and the third has $[4^{12}]$.

3.6 The plateau: 64 maximal sets, and the moves between them

Call $v \notin S$ *free* if $S \cup \{v\}$ is still 60° -free, and define the *tightness* of v to be $\#\{s \in S : \langle v, s \rangle = 16\}$. A set is maximal iff it has no free vertex; and since a $(k, k+1)$ -swap needs a vertex of tightness $\leq k$, a minimum tightness of 4 rules out all swaps with $k \leq 3$ at a stroke.

Six pairwise commuting plateau moves exist — four of size (12, 12) and two of size (80, 80) — all simultaneously applicable, generating a 6-cube of $2^6 = 64$ independent 496-element sets. Every one of the 64 has: zero free vertices, no vertex of tightness 1, 2 or 3, minimum tightness exactly 4 with exactly 80 tightness-4 vertices, and the identical Gram histogram. They fall into exactly 5 tightness-fingerprint

classes of sizes 16/16/16/8/8. Each is independently a certificate for $\alpha \geq 496$. [CERTIFIED-ATTACHED]
— the 64 vector lists are attached.

The 64 are pairwise inequivalent under Co_0 , and fall into 8 isometry classes of size 8: no plateau move is realised by a Leech automorphism, and seven of the 63 non-trivial sets are congruent copies of the record in $\mathbb{R}^{24}$ that no element of Co_0 reaches. The proof is an exhaustive individualisation-refinement enumeration of all Gram-graph isomorphisms (no pruning can miss one, because the colours are shared across the disjoint union), each extended to its unique orthogonal map of $\mathbb{R}^{24}$ and tested for preservation of Λ on all 196560 minimal vectors; five positive controls, including images overlapping the record in as few as four vectors, are correctly recognised. [CERTIFIED-HELD]

The moves are channel rewirings. This is the structurally decisive statement. The 80 tightness-4 vertices of every plateau set are 20 duad orbits confined to the three frame channels: the full *complementary* perfect matching of one frame channel (12 orbits) plus four one-pair orbits in each of the other two. Every conflict with S is a rook-row collision — a tightness-4 orbit conflicts with exactly the two S -orbits sharing one of its two rows, which is exactly why the minimum tightness is $4 = 2 \text{ orbits} \times 2 \text{ vectors}$. There are no cross-channel conflicts anywhere in S together with its tightness-4 vertices.

Each (12, 12) move is an alternating 6-cycle row-rotation inside one frame channel, and the four of them compose to a frame-for-frame swap. Each (80, 80) move removes *all* 20 orbits of a *pair* of channels $\{c, c \oplus c_F\}$ and installs a fresh 20-orbit wiring. In every case the 45-cell usage profile is *exactly preserved*. The moving parts of the whole 64-set plateau are precisely the three pairs of channels swapped by the translation $c \mapsto c \oplus c_F$ in $\text{PG}(3, 2)$; the nine one-pair channels and the empty channel never move.

The consequence is worth stating in isolation: *the cell profile is an invariant of the entire plateau, not merely of the record*. Local search never changes how much each cell carries, only which matching realises it. Any 497-element duad set must therefore break the profile itself.

3.7 Refuted hypotheses

These are the tempting guesses that turned out to be false. All [CERTIFIED-HELD], all verified negatives.

1. **The set is not all duads.** It is 496 of 11520; per block pair it holds 208 or 144 of 3840. The duads are very far from independent — the duads on a single line form a 16-clique, and there are 20160 rook conflicts, 322560 same-pair cross-channel conflicts and 23040 cross-pair same-channel conflicts among duad orbits. (Cross-pair cross-channel conflicts number exactly 0.)
2. **The set is not the set of minimal vectors of the lattice it generates.** The lattice $\langle S \rangle_{\mathbb{Z}}$ has rank 24 and index 16 in Λ with quotient $(\mathbb{Z}/2)^4$, and it equals $\text{Fix}(t_1) + \text{Fix}(t_2) + \text{Fix}(t_3)$, the sum of the three BW_{16} pair lattices. Its minimal vectors number 12240 — all 720 monads and all 11520 duads. So 11744 minimal vectors of $\langle S \rangle$ lie outside S , including all 80 tightness-4 vertices. The quotient has exponent 2, so $2\Lambda \subset \langle S \rangle$ and membership is a mod- 2Λ condition.
3. **The class set is not a union of cosets.** The 248 antipodal classes of S occupy 248 *distinct* classes of $\Lambda/2\Lambda$ (forced: $x \pm y \in 2\Lambda$ would need $|x \pm y|^2/4 \geq 32$), spanning a 20-dimensional $\mathbb{F}_2$ -subspace. A complete test over all 247 candidate translations shows the stabiliser $\{t : \text{class set} + t = \text{class set}\}$ is trivial. (By contrast the *channel* set does have stabiliser of order 2, generated by the empty channel, and no channel class is itself a class of an S -vector.)
4. **The octad incidence of the frames is not the Golay code.** Fix a frame; each of the 224 classes of S outside it is orthogonal to exactly 8 of the frame's 24 classes, giving a 24-bit mask. Spanning these masks over $\mathbb{F}_2$ gives, for two of the three frames, a self-complementary $[24, 7]$ code — 128 words, weight distribution $\{0:1, 8:39, 12:48, 16:39, 24:1\}$ — and for the third a self-complementary $[24, 9]$ code with 512 words and weights $\{0:1, 4:12, 8:135, 12:216, 16:135, 20:12, 24:1\}$. The Golay code is $[24, 12, 8]$ with 4096 words and 759 octads; neither of these is it, and the third even has weight-4 words, which a minimum-distance-8 code cannot. (This is the same asymmetry as the Smith-form split above: two frames behave alike, the third does not.)
5. **The $248 = \dim E_8$ numerology is dead.** $496 = 2^4 \cdot 31$ is a perfect number and $248 = \dim E_8$, which invites a 31-fold or E_8 -Lie symmetry. But $|\text{Co}_0| = 8315553613086720000 = 2^{22} \cdot 3^9 \cdot 5^4 \cdot 7^2 \cdot 11 \cdot 13 \cdot 23$, and $31 \nmid |\text{Co}_0|$. By Lagrange no element of order 31, and no subgroup of order divisible by 31, exists in Co_0 ; no such symmetry of a Co_0 -stable object can exist. The natural factorisation is $496 = 4 \cdot 124$ with $124 = 3 \cdot 12 + 9 \cdot 8 + 2 \cdot 8$, and 248 is just $496/2$. The genuine E_8 content of the structure is real

but elsewhere: $\Lambda \cap B_i \cong \sqrt{2}E_8$, the 135 E_8 frames per block, and the E_8 root-line strongly regular graph above.

Two more negatives worth recording. The ± 16 -free two-point LP is *not* below 850.9 — it is exactly 9360/11, as Section 2.2 showed. And the 208 on the largest pair is not even maximal within its own pair: the single-pair duad graph is 252-regular, its ratio bound is about 282, there exist pair duads with zero conflicts against the 208, and the true single-pair optimum is 64 orbits = 256 vectors against the record’s 52 orbits = 208. It is the cross-pair constraints that pin the record down, not the within-pair ones.

4 Open questions, ranked

4.1 Is 496 maximum in the duad model?

The finite model is small and completely explicit: 2880 duad orbits plus 360 monad orbits, with the rook condition per cell, the $K_{8,8,8}$ matching condition per channel, the cross-channel rule of Section 3.3, and the plane-local monad rule. Maximise total weight. The record is 124 orbits; the question is whether 125 exists.

Tractability: the local structure is all proved. Within a single pair, the maximum on a channel subset $\mathcal{C}$ is exactly $8 \cdot (\text{largest PG}(3, 2)\text{-cap inside } \mathcal{C})$ — every single channel gives 8, every pair of channels 16, every *line* $\{c, c', c \oplus c'\}$ gives 16 (not 24), every non-collinear triple 24, every plane 32, the whole pair 64. Monads are provably scarce: the exact “kill function” $u(k) = 14, 26, 36, 44, 50, 54, 56, 56$ (the number of block lines destroyed by an independent k -set of one block’s monads, achieved inside a single plane) forces full-monad solutions down to weight $432 < 496$. The symmetry group of the reduced model acts on (channels, pairs) as the full $\text{GL}(4, 2) \times S_3$, order 120960, so any cap proved for one shape holds for all its images.

What blocks it is described in §4.4. My estimate: this is the most likely of the four to fall, and the right tool is a combinatorial argument on the rook/PG(3, 2) model rather than a solver — the object is now genuinely small.

4.2 Must a maximum set sit on a Turyn frame at all?

Every 496-element set anyone here has produced or been given — the record, the 64-set plateau, a thousand further sets from GPU search — is a duad set on *some* Turyn frame. Nothing proves that a maximum set must be. If the duad question above were settled at 496, this would reduce “496 is optimal” to “a maximum set admits a blockwise 2^3 ”. The only evidence is the subgroup sweep of Section 2.7: the best set invariant under any group other than the record’s own stabilisers is 402. That is weak evidence and should be treated as such. I have no tractability estimate for this one; it is the question I would most like an idea about.

4.3 The four-point bound

Shovel-ready and fully scoped (Section 2.5): orbit data complete, largest block 4107, 64–128 GB, two to four weeks including a sparse-integer rewrite of the certificate path, expected landing 820–830. This narrows the gap but almost certainly does not close it. If either of you has access to stronger SDP infrastructure, this is the item that is simply waiting for a machine.

4.4 The Turyn-symmetric slice: $496 \leq \text{opt} \leq 720$

Restrict to sets invariant under the blockwise sign group. The lower end is the record (an independent union of 124 orbits). The upper end is $4 \times 15 \times 12 = 720$, from the per-channel matching cap alone, reproduced independently by a clique-cover LP and by a constraint solver. [EVIDENCE] for the upper end: it rests on solver verdicts, not on an extracted exact certificate.

The obstruction is sharp and worth stating precisely, because it explains why nothing moves.

The flat profile. Put exactly 4 orbits in *each* of the 45 cells and use no monads. This profile satisfies every quickly-provable local cap: the rook cap ($4 \leq 8$), the $K_{8,8,8}$ matching cap ($4 + 4 + 4 = 12$, exactly saturating), every proved per-pair line cap (≤ 16), plane cap (≤ 32) and pair total (≤ 64 ; a flat pair carries 60), and the monad kill cut vacuously. It pays $15 \cdot 12 \cdot 4 = 720$.

So every linear or Lagrangian relaxation stalls at exactly 720 — not approximately, exactly — while more than ten hours of exact anytime search never found a 497. The record’s own profile is nothing like flat: it is the highly uneven 4/4/4, 8/0/0, 0/4/4, 0/0/0 table of Section 3.4, 56 orbits short of the flat 180. The gap between what a pair can actually do and what a relaxation can prove is enormous where it matters: with every channel capped at k , the best witness in one pair for $k = 2, 3, 4$ is 22, 27, 36 duads while the best proved bound is 30, 45, 60. A pair genuinely cannot spread thin, and no relaxation tried can see it. Spectral methods do not help either: the duad-only conflict graph is 254-regular with $\lambda_{\min} = -18$ exactly, giving a ratio bound of 190.6 orbits = 762.4 weight, and a symmetry-reduced weighted ϑ comes out around 1031, worse than the clique LP.

Moving below 720 therefore requires refuting a subproblem combinatorially. The natural targets, with the bound each would certify: pinning the $k = 2$ per-pair cap at its witness value gives ≤ 630 ; adding $k = 3$ gives ≤ 588 ; a plane subproblem across all three pairs gives ≤ 560 . A refutation of the full ≥ 497 instance would be the theorem itself.

One correction to record here, because it is the kind of thing that propagates: an earlier claim that nine simultaneously perfect channels are infeasible was a mislabelled timeout. Eight simultaneously perfect channels provably exist (on an affine complement of a plane); “exactly eight” is open. The record has three perfect channels not because four is impossible — four, five, six, seven and eight are all feasible — but as an optimisation trade-off.

5 Method

The work was carried out by AI agents operating under my direction, with a two-verifier rule for every headline claim: no set, family or bound was recorded until two independent implementations, sharing no code path and regenerating their inputs separately, agreed on it. All certified statements are computed in exact integer or rational arithmetic — no floating-point verdict is ever the last word — and the headline claims were reimplemented independently in a second language. That rule earned its keep at least once: a driver bug produced a run of spurious 497–568 candidates, both verifiers rejected every one of them, and no false record was logged.